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region. Endodermis is indistinguishable from cortical cells. Vascular region is surrounded by pericycle. A ring consists of five or six vascular bundles are present. Vascular bundles are conjoint, collateral, open and endarch. Pith and medullary rays are present. Secondary growth is present and annual rings are formed.

T.S. of needle or foliage leaf

The internal structure of needle shows xerophytic adaptations. In cross section the outline appears more or less triangular and is divided into epidermis, mesophyll and vascular bundles. The epidermis is single layered and possesses thick cuticle and sunken stomata.Epidermis is followed by a few layers of sclerenchymatous hypodermis. It is interrupted by sub-stomatal cavities (Figure 2.51).

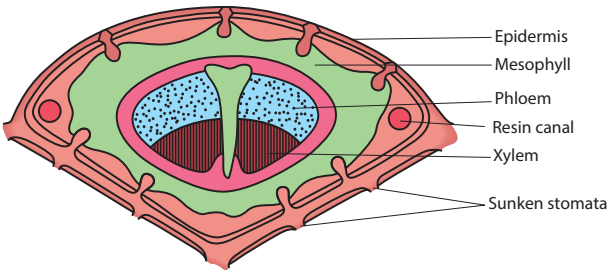


Figure 2.51: T.S. of *Pinus* needle leaf

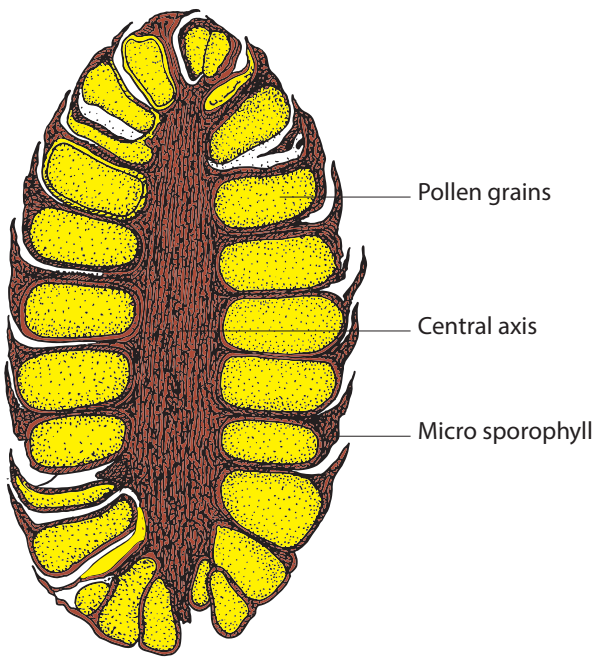
Mesophyll is not differentiated into palisade and spongy parenchyma. Thin walled cells with chloroplasts are present. The cells are peculiar with numerous small infoldings which project into the cavities. The infoldings increase the photosynthetic area of the needle leaves Resin canal is present in the mesophyll. A single layered endodermis separates the vascular region from the cortex. A multilayered pericycle containing starch is present. Two types of specialised cells called **albuminous cells** and **tracheidal cells** are present. The former helps to pass substances from the mesophyll to the phloem while the latter helps in water conduction and constitutes transfusion tissue. Two vascular bundles are present. They are separated by sclerenchyma tissue. The Vascular bundles are conjoint, collateral and open.

Reproduction

Pinus is heterosporous and produces two types of spores called. microspores and megaspores. The plants are monoecious. Both male and female cones or strobili develop on the different branches of the same plant (Figure 2.52).



a) *Pinus* - A twig with male cones



b) L.S. of male cone